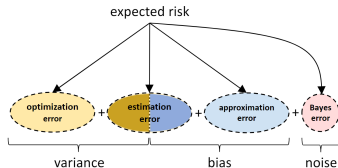


## Bias-Variance 2:

## Approximation and Estimation error



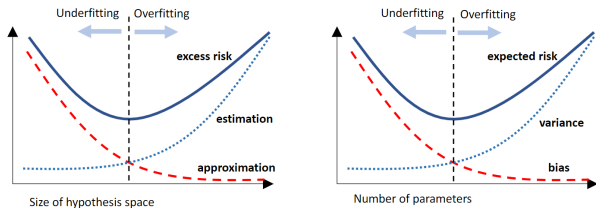
## Learning goals

- Decomposing excess risk
- Into estimation, approx. and optim. error

- BV decomp often confused with related (but different) decomp:

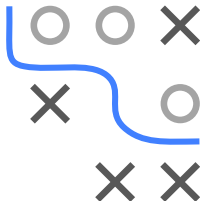
$$\underbrace{\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}_{all}}^*)}_{\text{excess risk}} = \underbrace{\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}}^*)}_{\text{estimation error}} + \underbrace{\mathcal{R}(f_{\mathcal{H}}^*) - \mathcal{R}(f_{\mathcal{H}_{all}}^*)}_{\text{approx. error}}$$

- Both commonly described using same figure and analogies



► [Click for source](#)

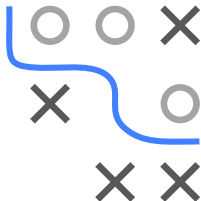
- BV decomp. only holds for certain losses, above is universal



- Approx. error is a structural property of  $\mathcal{H}$
- Estimation error is random due to dependence on data in  $\hat{f}$
- Estimation error occurs as we choose  $f \in \mathcal{H}$  with limited train data minimizing  $\mathcal{R}_{\text{emp}}$  instead of  $\mathcal{R}$
- Knowing  $\hat{f}_{\mathcal{H}} \in \arg \inf_{f \in \mathcal{H}} \mathcal{R}_{\text{emp}}(f)$  assumes we found a global minimizer of  $\mathcal{R}_{\text{emp}}$ , which is often impossible (e.g. ANNs)
- In practice, optimizing  $\mathcal{R}_{\text{emp}}$  gives us “best guess”  $\tilde{f}_{\mathcal{H}} \in \mathcal{H}$  of  $\hat{f}_{\mathcal{H}}$
- Can now decompose its excess risk finer as

$$\underbrace{\mathcal{R}(\tilde{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}_{\text{all}}}^*)}_{\text{excess risk}} = \underbrace{\mathcal{R}(\tilde{f}_{\mathcal{H}}) - \mathcal{R}(\hat{f}_{\mathcal{H}})}_{\text{optim. error}} + \underbrace{\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}}^*)}_{\text{estimation error}} + \underbrace{\mathcal{R}(f_{\mathcal{H}}^*) - \mathcal{R}(f_{\mathcal{H}_{\text{all}}}^*)}_{\text{approx. error}}$$

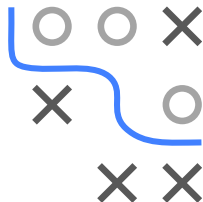
- NB: Optim err. can be  $< 0$ , but  $\mathcal{R}_{\text{emp}}(\tilde{f}_{\mathcal{H}}) \geq \mathcal{R}_{\text{emp}}(\hat{f}_{\mathcal{H}})$  always



- We can further decompose estimation error more finely by defining the *centroid* model or “systematic” model part
- For  $\hat{f}_{\mathcal{H}} \in \arg \min_{f \in \mathcal{H}} \mathcal{R}_{\text{emp}}(f)$  centroid model under L2 loss is mean prediction at each  $x$  over all  $\mathcal{D}_n$ ,  $f_{\mathcal{H}}^{\circ} := \mathbb{E}_{\mathcal{D}_n \sim \mathbb{P}_{xy}^n} [\hat{f}_{\mathcal{H}}]$
- With  $f_{\mathcal{H}}^{\circ}$ , can decompose expected estimation error as

$$\underbrace{\mathbb{E}_{\mathcal{D}_n \sim \mathbb{P}_{xy}^n} [\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}}^*)]}_{\text{expected estimation error}} = \underbrace{\mathbb{E}_{\mathcal{D}_n \sim \mathbb{P}_{xy}^n} [\mathcal{R}(\hat{f}_{\mathcal{H}}) - \mathcal{R}(f_{\mathcal{H}}^{\circ})]}_{\text{estimation variance}} + \underbrace{\mathcal{R}(f_{\mathcal{H}}^{\circ}) - \mathcal{R}(f_{\mathcal{H}}^*)}_{\text{estimation bias}}$$

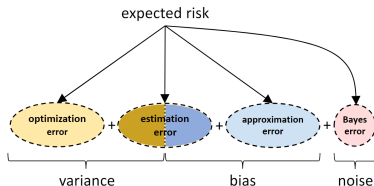
- Estimation bias measures distance of centroid model to risk minimizer over  $\mathcal{H}$
- Estimation var. spread of ERM around centroid model induced by randomness due to  $\mathcal{D}_n$



- Can now connect derived quantities back to bias and variance
- Bias is not only approx. error and variance is not estimation error
- Many details skipped here, see paper!

$$\text{bias} = \text{approximation error} + \text{estimation bias}$$

$$\text{variance} = \text{optimization error} + \text{estimation variance}$$



► Click for source

- **NB:** For special case of LM and L2 loss, we have very small optim / numerical error and estimation bias; so both decomp agree

